

3.0 Points

Which is not true of the Advanced Encryption Standard (AES)?

- ☐ A. It is a block symmetric cipher
- ☐ B. It uses the algorithm Rijndael
- ☐ C. It was developed to replace DES
- ☒ D. It uses key sizes of 64, 128 and 192
- ☐ E. It's faster than RSA

AES key sizes may be any multiple of 32 bits, both with a minimum of 128 and a maximum of 256 bits

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3.0 Points

Keys that create a repeating pattern are known as:

- ☐ A. repeating keys
- ☐ B. inferior algorithms
- ☒ C. weak keys
- ☐ D. cipher abnormalities (CA)
- ☐ E. structure algorithms

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3.0 Points

In 1976, Diffie and Hellman introduced what cryptography technology?

- ☐ A. 256-bit key encryption capabilities
- ☐ B. Digital signatures
- ☐ C. Quantum cryptography
- ☒ D. Electronic key distribution
- ☐ E. Symmetric key encryption

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3.0 Points

A fixed-length value used as a message fingerprint is called a:

- ☐ A. Digital signature
- ☐ B. Message value
- ☐ C. MAC
- ☒ D. Hash value
- ☐ E. Cipher

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3.0 Points

This is a trial and error method used to decode encrypted data through exhaustive effort rather than employing intellectual strategies:

- ☐ A. serendipity
- ☐ B. chaffing and winnowing
- ☒ C. brute force cracking
- ☐ D. rubber hose
- ☐ E. cryptanalysis

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3.0 Points

The inclusion of a secret message in otherwise unencrypted text or images is called

- ☐ A. eye-in-hand system
- ☒ B. steganography
- ☐ C. cryptography
- ☐ D. spoof
- ☐ E. masquerade

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3.0 Points

An attacker who has access to a large section of ciphertext, then determines which part of it is to be decrypted, and ultimately has access to the resulting plaintext has performed what type of attack?

- ☐ A. Chosen-key
- ☒ B. Chosen-ciphertext
- ☐ C. Known-plaintext
- ☐ D. Adaptive chosen-plaintext
- ☐ E. Chosen-plaintext
- ☐ F. Ciphertext-only

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3.0 Points

This is a mode of operation for a block cipher, with the characteristic that each possible block of plaintext has a defined corresponding ciphertext value and vice versa

- ☐ A. footprinting
- ☐ B. watermark
- ☐ C. Cipher Block Chaining
- ☒ D. Electronic Code Book
- ☐ E. hash function

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3.0 Points

In password protection, this is a random string of data used to modify a password hash:

- ☐ A. padding
- ☒ B. salt
- ☐ C. MAC
- ☐ D. dongle
- ☐ E. bypass
- ☐ F. sheepdip

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3.0 Points

A secret key that's used for data encryption only one time is called a:

- ☐ A. MAC key
- ☒ B. Session key
- ☐ C. Key exchange
- ☐ D. Asymmetric key
- ☐ E. Public key

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3.0 Points

Julius Caesar constructed his own encryption method to hide data during transmission or while being stored. The Caesar Cipher works in which of the following ways?

- ☐ A. Letters are replaced by randomly generated numbers
- ☐ B. Replaces letters of the alphabet with letters 13 characters beyond it.
- ☐ C. Letters are randomly scrambled
- ☒ D. Each letter in the alphabet is replaced with a letter three places beyond it.
- ☐ E. The message itself consists of clues to recover from different places in the physical world.

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3.0 Points

The encryption algorithm that has replaced the Data Encryption Standard (DES) - and later Triple DES - as the new standard encryption algorithm is:

- ☐ A. 3DES
- ☐ B. Kerberos
- ☐ C. IPsec
- ☐ D. Blowfish
- ☐ E. RSA
- ☒ F. Rijndael

The Rijndael algorithm has been selected as the Advance Encryption Standard (AES) to replace 3DES. AES is modified version of Rijndael algorithm.

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3.0 Points

Which is the 128-bit algorithm developed by IBM that was accepted for the DES standard?

- ☐ A. None of these options
- ☐ B. Data Encryption Algorithm
- ☐ C. RSA
- ☐ D. Rijndael
- ☐ E. Skipjack
- ☒ F. Lucifer

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3.0 Points

The coin-flipping protocol between Alice and Bob is based on:

- ☐ A. Using a large prime number
- ☐ B. A MD5 hash function
- ☒ C. A magic function f that is injective
- ☐ D. 50% probability of winning when Bob's flip of a coin
- ☐ E. Trust between the two parties

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3.0 Points

A stream cipher type is

- ☐ A. DES
- ☒ B. RC4
- ☐ C. AES
- ☐ D. RSA
- ☐ E. Caesar

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3.0 Points

Which of these methods represent a pseudo-random numbers generator (PRNG's):

- (1) Yarrow
- (2) Blum Blum Shub
- (3) ANSI X9.17
- (4) Linear congruence method
- (5) Linear feedback shift register - LFSR

- ☐ A. 1+2+3+4+5
- ☐ B. 1+2+3+5
- ☐ C. 1+2+5
- ☐ D. 1+2+3+4
- ☐ E. 1+2+3
- ☒ F. 1+2+4+5

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3.0 Points

Which of the following algorithms are NOT used for implementation of a hash function?

- ☐ A. RIPEMD-160
- ☐ B. Tiger
- ☐ C. MD5
- ☐ D. MD2
- ☒ E. EG - El Gamal

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3.0 Points

It is given the value $N = 247$ and the set of integer values $S = \{1, 2, 3, \dots, 189, 190\}$.

Using the backpack problem determine all possible subsets of items from S which represent solutions for N and indicate which one of these solutions is present here:

- ☐ A. $\{10, 25, 30, 42, 47, 99\}$
- ☐ B. $\{10, 20, 30, 40, 47, 101\}$
- ☒ C. $\{10, 20, 30, 43, 44, 100\}$
- ☐ D. The time allocated for the exam is NOT enough to solve this problem
- ☐ E. $\{1, 29, 30, 40, 47, 101\}$

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3.0 Points

The RSA (Rivest-Shamir-Adleman) cipher is based on

- ☒ A. Factorization problem
- ☐ B. Elliptic curves theory
- ☐ C. The discrete logarithms problem
- ☐ D. none of these options
- ☐ E. The backpack or knapsack problem

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3.0 Points

ElGamal encryption scheme is based on

- ☐ A. none of these options
- ☐ B. The backpack or knapsack problem
- ☐ C. Factorization problem
- ☒ D. The discrete logarithms problem
- ☐ E. Elliptic curves theory

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3.0 Points

Which one of the following cryptographic algorithms can be used both for encryption and electronic signature of a message:

- ☐ A. DES – Data Encryption Standard
- ☐ B. DSS - Digital Signature Standard
- ☐ C. FEAL - Japanese Fast Data Encryption Algorithm (FEAL-1)
- ☒ D. RSA - Rivest Shamir Adleman
- ☐ E. none of these options
- ☐ F. AES - Advanced Encryption Standard - Rijndael

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3.0 Points

Indicate the correct statement that describes one of the principles of cryptography:

- ☐ A. A best practice is to keep the algorithm secret
- ☐ B. Used algorithms require a high computational effort to verify the solution
- ☒ C. None of the above statements
- ☐ D. Used algorithms have solutions that can be determined in a certain period of time
- ☐ E. Does not implement algorithms that are defined based on NP complex mathematical models

Principles of cryptography:

1. Encryption
2. Authentication
3. Integrity
4. Non-repudiation

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3.0 Points

P-Box mechanism used by modern cryptographic algorithms represents the result of using and analyzing one of the next ciphers:

- ☐ A. mono-alphabetic substitution
- ☐ B. polygram substitution
- ☐ C. poly-alphabetic substitution
- ☐ D. none of these options
- ☒ E. transposition cyphers

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3.0 Points

MD5 hash function generates values with a size equal to:

- ☐ A. 256 bits
- ☐ B. 192 bits
- ☐ C. 16 bits
- ☒ D. 128 bits
- ☐ E. 64 bits

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3.0 Points

Which statement is false?

- ☐ A. ANSI X9.17 is NOT an encryption algorithm
- ☐ B. ANSI X9.17 PRNG is based on triple-DES encryption
- ☐ C. Blum Blum Shub PRNG is based on the factorization problem
- ☒ D. Blum Blum Shub does NOT pass next-bit test
- ☐ E. Yarrow is one of the most popular PRNG

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3.0 Points

Select the correct answer regarding DES (Data Encryption Standard)

- ☐ A. It has known weak keys
- ☐ B. none of these options
- ☒ C. It is a symmetric block cipher
- ☐ D. It uses 8 byte blocks
- ☐ E. It is faster than RSA

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Question 5 of 30

3.0 Points

DES has gone through different generations: DES, Double DES, 3DES. 3DES is how much stronger than DES from a cryptographic view point?

- ☒ A. 2 exponent 56 = 2^{56}
- ☐ B. 3
- ☐ C. 2 exponent 3 = 2^3
- ☐ D. 2 exponent 64 = 2^{64}
- ☐ E. 192

Question 9 of 30

3.0 Poi

Which of the following is not true of RSA?

- ☐ A. Developed at MIT by Ron Rivest, Adi Shamir, and Leonard Adleman
- ☒ B. Was accepted as the new Advanced Encryption Standard in the late 1990s
- ☐ C. It is using large prime numbers of at least 1024 bits
- ☐ D. Can be used for encryption and digital signatures
- ☐ E. Can be used for key exchange

Question 10 of 30

3.0 Points

In a 64-bit Data Encryption Standard (DES) key, how many bits make up the true key and how many bits make up parity?

- ☐ A. 48 and no parity
- ☐ B. 48 + 16 for parity
- ☐ C. 16 + 48 for parity
- ☐ D. 64 and no parity
- ☐ E. 56 and no parity
- ☒ F. 56 + 8 for parity

Question 13 of 30

3.0 Points

This was commonly used in cryptography during World War II

- ☐ A. RSA
- ☐ B. van Eck phreaking
- ☐ C. tunneling
- ☒ D. one-time pad
- ☐ E. personalization
- ☐ F. AES

Question 18 of 30

3.0 Points

What types of cryptographic services are defined by X.800 standard ?

(1) Authentication

(2) Access Control

(3) Data Confidentiality

(4) Data Integrity without Recovery

(5) Nonrepudiation

☐ A. 1+3+5

☐ B. 2+3+4+5

☒ C. 1+2+3+4+5

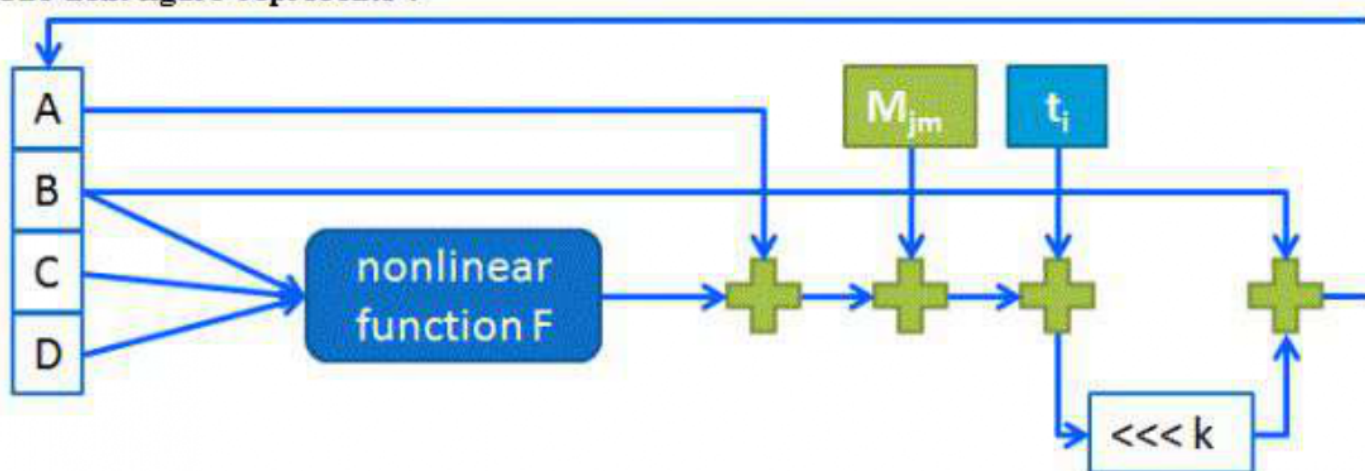
☐ D. 1+2+4+5

☐ E. 1+2+3+5

Question 21 of 30

3.0 Points

The next figure represents ?

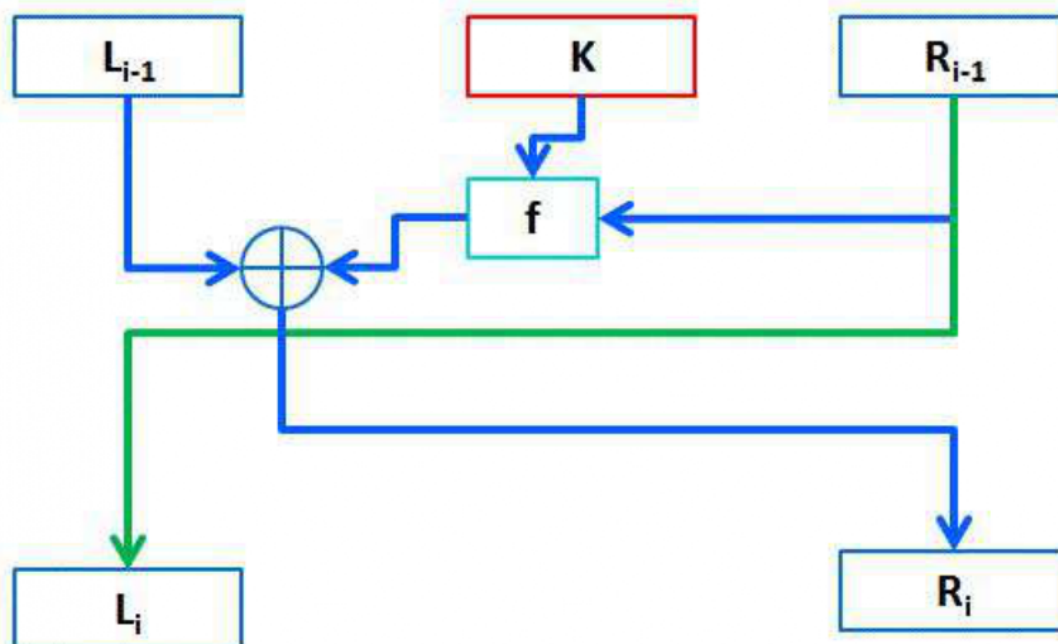


- ☒ A. Scheme of one round from the MD5 hash algorithm;
- ☐ B. Scheme of the hash algorithm MD5;
- ☐ C. The Euler function used in RSA algorithm;

Question 22 of 30

3.0 Points

Feistel network shown in next figure is often encountered in:



- ☐ A. hash functions
- ☒ B. symmetric-key cryptographic algorithms
- ☐ C. none of these options
- ☐ D. public key distribution systems
- ☐ E. asymmetric-key cryptographic algorithms (public key)

Question 24 of 30

3.0 Points

It is NOT a secure solution to generate random numbers used in by cryptographic algorithms:

- ☐ A. Linear feedback shift register method
- ☒ B. C rand function
- ☐ C. Recording pressed keys within a period of time
- ☐ D. Global geo-position of a lightning
- ☐ E. Recording cursor position within a period of time

Question 26 of 30

3.0 Points

Which of the listed algorithms can NOT be used to produce electronic signatures:

- ☐ A. EG-EC - elliptic curves El Gamal;
- ☒ B. none of these options
- ☐ C. EG - El Gamal;
- ☐ D. RSA - Rivest Shamir Adleman;
- ☐ E. DSS - Digital Signature Standard;

Question 28 of 30

3.0 Points

Indicate which of the following statements is NOT an objectives of a modern and secure cryptographic system:

- ☐ A. Ensures total or partial confidentiality
- ☐ B. Ensure data integrity
- ☒ C. Implement cryptographic algorithms developed in secret by teams of specialists
- ☐ D. Non-repudiation of messages
- ☐ E. Ensures data integrity
- ☐ F. Authenticates communication parties

Different types of algorithms use different types of mathematics. The more complex the mathematics are, the more resources that are required for computation. Which asymmetric algorithm is the most efficient requiring the fewest resources?

A. RSA

B. ECC

C. Blowfish

D. IDEA

CORRECT ANSWER - B. ECC

Elliptic Curve Cryptosystems (ECC) is the most efficient of the asymmetric algorithms. It uses elliptic curve properties to combine group and rule information, which requires fewer resources than the other methods.

Which of the following algorithms was not considered by NIST when determining the algorithm that was to be used for the new AES in 1997?

A. Twofish

B. MARS

C. Rijndael

D. El Gamal

CORRECT ANSWER - D. El Gamal

The Advanced Encryption Standard (AES) uses a symmetric block algorithm. El Gamal is a public key (asymmetric) encryption system, and thus was not considered for the advanced standard. El Gamal is similar to RSA; however, it utilizes discrete logarithms in a finite field.

Which key knows the trapdoor that allows for decryption to take place?

- A. Session key
- B. Public key
- C. Private key**
- D. Asymmetric key

CORRECT ANSWER -

A trapdoor one-way function applies the concept of factoring numbers into their original prime numbers. Public keys encrypt a message with a built-in, one-way function. This function is referred to as a one-way function because it's relatively simple to encrypt, but much more difficult to decrypt without knowing the correct trapdoor. A private key, however, knows the code of the trapdoor, and is able to decrypt the message.

The Hash of Digital signatures are encrypted with the

- ☒ receiver's public key
- ☒ sender's public key
- ☒ sender's private key
- ☒ receiver's private key

What would be the best hash algorithm to store a password?

- ☒ SHA1
- ☒ AES
- ☒ MD5
- ☒ SHA256

What are the services provided by asymmetric cryptography? Select all that apply.

- ☒ Availability
- ☒ Key exchange
- ☒ Privacy
- ☒ Non-repudiation
- ☒ Integrity
- ☒ Authentication

Digital signatures provide

- ☒ integrity
- ☒ authenticity
- ☒ non repudiation
- ☒ all choices

Block ciphers are which type of encryption?

- ☒ iii) Both i) & ii)
- ☒ i) Symmetric
- ☒ ii) Asymmetric
- ☒ iv) None of these

Which one of these is a popular key exchange algorithm:

- ☒ 3 DES
- ☒ DES
- ☒ AES
- ☒ Diffie-Hellman

What security requirement does hashing with digital signatures fulfill?

- ☒ Authentication
- ☒ Non-repudiation
- ☒ Integrity
- ☒ Availability

Which asymmetric algorithm uses a 160-bit key which is almost equivalent to the protection given by 1024-bit RSA key?

- ☒ 3-DES
- ☒ RSA
- ☒ ECC
- ☒ ElGamal

The algorithm type that generates a key stream which is XOR'd with plaintext for encryption purposes is the what?

- ☒ Initialization vector
- ☒ Stream Cipher
- ☒ Block cipher
- ☒ Key stream generator

Which is a set of ciphers where the first one creates diffusion and the second one creates confusion?

- ☒ Transposition, AES
- ☒ One-time pad, Substitution
- ☒ Transposition, Caesar
- ☒ Transposition, Substitution

Which of these attacks describes a known plaintext attack?

- ☒ The session key is intercepted and used against a later encrypted session
- ☒ An attacker intercepts messages between two parties on the network and modifies the original message
- ☒ An attacker encrypting known plaintext with every possible key to decrypt the corresponding ciphertext
- ☒ The attacker has obtained a ciphertext and plaintext for the past messages which they use to decipher new messages

Which of the following modes of DES is the weakest?

- ☒ OCB
- ☒ ECB
- ☒ BCE
- ☒ OEB

Substitution ciphers may be monoalphabetic or polyalphabetic. When using a monoalphabetic substitution cipher, a single alphabet is used to encrypt what?

- ☒ Each bit
- ☒ Each character
- ☒ Each character block
- ☒ The entire plain text message

An attacker was questioned about the attack vector he used to decrypt an encrypted data file, and he confessed that he analyzed the messages based on encrypted message length. Which cipher was he trying to attack here?

- ☒ None of these
- ☒ Symmetric
- ☒ Stream
- ☒ Block